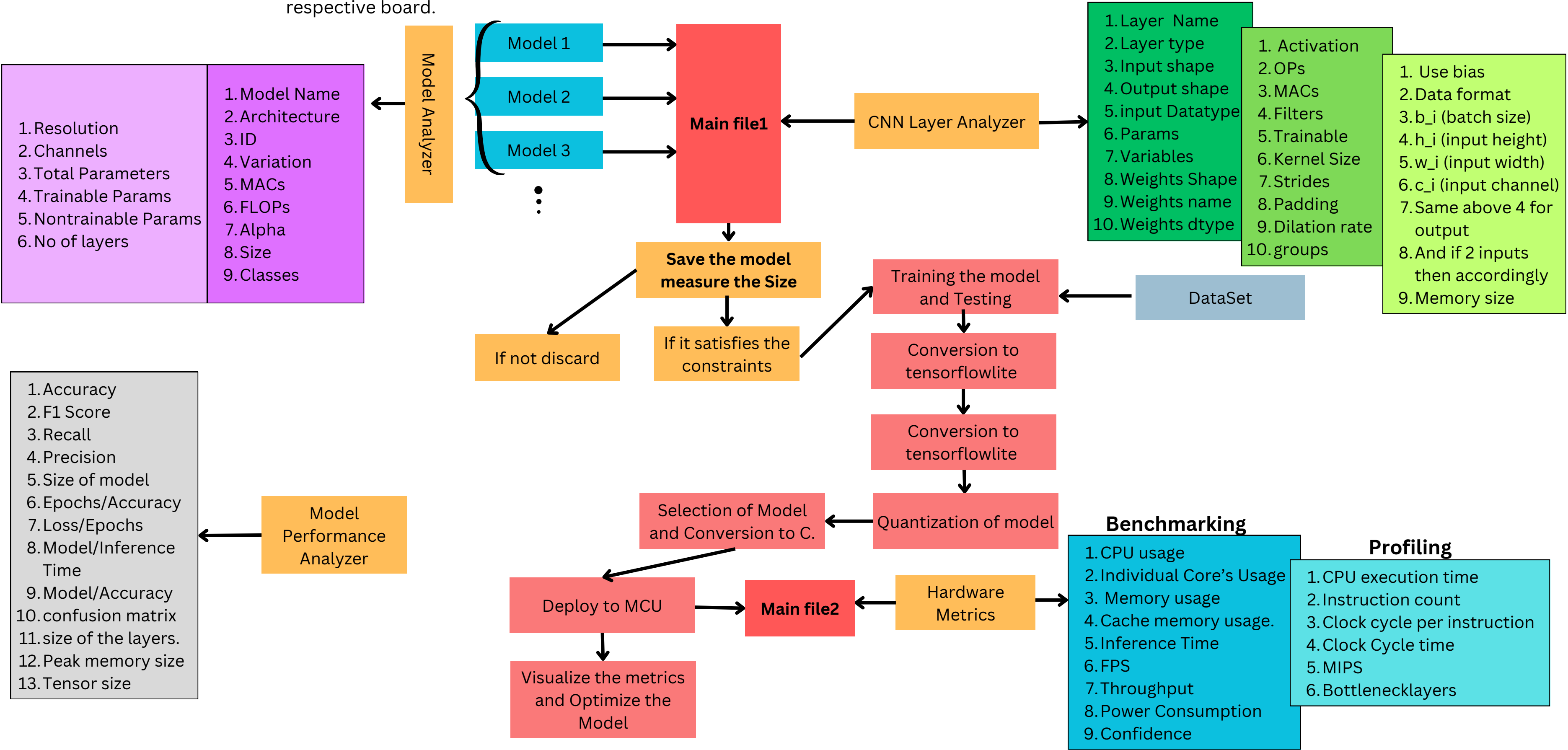


EdgeAI Metrics Measurement Framework

Objective: The Framework for testing and extracting metrics for EdgeAI models to decide the model for the respective board.



PROGRAMMING LANGUAGE AND ITS EFFECT ON THE PERFORMANCE

**CNN IN
PYTHON**

**CNN IN
PYTHON
CONVERTED
TO C**

CNN IN C

**ASSEMBLY
LANGUAGE
CODING**

**METRICS TO BE
CALCULATED**



1. Execution Time
2. Memory Usage
3. FLOPS
4. Energy Consumption
5. Model Accuracy
6. Scalability
7. Latency
8. Instruction Count
9. Parallelization and Hardware Utilization

Main file 1

For Model Training and conversion

1. Model.py- All the model architectures are defined.
2. Model analyzer.py- Model metrics are collected and csv generated.
3. utils.py- Common utilization and common helping fuctions.
4. Saved model- Temporarily storing the model.
- 5.

Main file 2

For Deployment on the board

1. Model.cc- Source file containing binary model data
2. Model.h- Header file (contains declaration)
3. Benchmark - Measures
 - * Inference time
 - * CPU Usage
 - * Invidual Core Usage
 - * Memory Tracking
 - *FPS= througput

Model Analyzer- Glossary

When designing and deploying deep learning models, understanding their properties is crucial for optimization, performance, and resource management. This glossary provides a detailed explanation of the key parameters used to evaluate and compare models in terms of efficiency, computational complexity, and deployment feasibility.

1 Resolution

- The input image size (e.g., 224×224, 128×128).
- Defines how much detail the model can capture.
- Larger resolutions capture more details but increase computational cost.

2 Channels

- Number of channels in the input image.
 - 1 for grayscale images.
 - 3 for RGB images.
- Affects the number of parameters in the first layer.

3 Total Parameters

- The sum of trainable and non-trainable parameters.
- Affects model size and memory footprint.

4 Trainable Params

- Parameters updated during training (weights and biases).
- More trainable parameters generally mean higher learning capacity, but also more computational cost.

5 Non-trainable Params

- Fixed parameters, usually in pretrained layers (e.g., frozen layers in transfer learning).
- Helps reduce training time and leverage existing knowledge.

6 No. of Layers

- The total number of layers, including convolutional, pooling, dense, etc.
- Deep models capture complex features, but also require more computation and memory.

7 Variation

- Refers to different versions or modifications of the model (e.g., MobileNetV2-1.0 vs MobileNetV2-0.75).
- Different variations have trade-offs in accuracy vs computational efficiency.

8 MACs (Multiply-Accumulate Operations)

- The number of multiplication-accumulation operations required per forward pass
- Helps estimate processing power requirements for deployment.

9 FLOPs (Floating Point Operations Per Second)

- The total number of mathematical operations (multiplication + addition).
- Measures computational complexity and determines hardware requirements.

10 Alpha (α)

- A scaling factor for model width in MobileNet architectures.
- Controls the number of channels in each layer:
 - $\alpha = 1.0 \rightarrow$ Full-sized model
 - $\alpha = 0.75 \rightarrow$ 75% of the channels
 - $\alpha = 0.5 \rightarrow$ 50% of the channels
- Allows trade-offs between accuracy and efficiency, useful for mobile & edge AI.

11 Size

- The total size of the model file (measured in KB).
- Affects storage, loading time, and deployment feasibility on different devices.

12 Classes

- The number of output classes.
- Defines the number of categories the model can classify.

Hardware Analyzer- Glossary

Once the trained model converted to C array and passed on to deploy on the Hardware. The main file consists of

- Preprocessing code for the input- Resize + quantize image input.
- Load the model
- Inference loop- Benchmark File.
- Post Processing- Confidence.

Interpreter file

1. Operations
2. Tensors
3. data type
4. Shape
5. Bias

TODO:

- 1. Custom Profiler.**
- 2. Convert Everything to c.**